OR4023 (Optimization) Homework 5

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Question 1. Prove that a matrix $\mathbf{C}$ is symmetric and of rank-one if and only if $\mathbf{C} = \gamma w w^T$ for some scalar γ and unit vector w.

Solution. We prove this statement in two directions.

Part 1: ($\Rightarrow$) If $\mathbf{C} = \gamma \mathbf{w}\mathbf{w}^T$ for some scalar γ and unit vector $\mathbf{w}$, then $\mathbf{C}$ is symmetric and rank-one.

1. **Symmetry:** We check if $\mathbf{C}^T = \mathbf{C}$.

$$\mathbf{C}^T = (\gamma \mathbf{w}\mathbf{w}^T)^T = \gamma(\mathbf{w}\mathbf{w}^T)^T = \gamma(\mathbf{w}^T)^T\mathbf{w}^T = \gamma \mathbf{w}\mathbf{w}^T = \mathbf{C}.$$

   Thus, $\mathbf{C}$ is symmetric.

2. **Rank-one:** Let $\mathbf{w}$ be an $n \times 1$ vector. Since $\mathbf{w}$ is a unit vector, $\mathbf{w} \neq \mathbf{0}$. The matrix $\mathbf{C} = \gamma \mathbf{w}\mathbf{w}^T$ is an $n \times n$ matrix. Each column of $\mathbf{C}$ is a scalar multiple of the vector $\gamma \mathbf{w}$. Specifically, the j-th column of $\mathbf{C}$ is $(\gamma w_j)\mathbf{w}$. If $\gamma = 0$, then $\mathbf{C}$ is the zero matrix, which has rank 0. If $\gamma \neq 0$, since $\mathbf{w} \neq \mathbf{0}$, the vector $\gamma \mathbf{w}$ is non-zero. Thus, the column space of $\mathbf{C}$ is spanned by the single non-zero vector $\gamma \mathbf{w}$. Therefore, the dimension of the column space of $\mathbf{C}$ is 1, which means $\text{rank}(\mathbf{C}) = 1$. (The problem implies "rank-one", so we assume $\gamma \neq 0$.)

Part 2: ($\Leftarrow$) If $\mathbf{C}$ is symmetric and rank-one, then $\mathbf{C} = \gamma \mathbf{w}\mathbf{w}^T$ for some scalar γ and unit vector $\mathbf{w}$.

1. Since $\mathbf{C}$ has rank one, all its columns must be scalar multiples of a single non-zero vector, say $\mathbf{u}$. So, $\mathbf{C}$ can be written in the form $\mathbf{C} = \mathbf{u}\mathbf{v}^T$ for some non-zero vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$. (If $\mathbf{C}$ is the zero matrix, its rank is 0, not 1. So $\mathbf{C}$ is not the zero matrix).

2. Since $\mathbf{C}$ is symmetric, $\mathbf{C} = \mathbf{C}^T$. Therefore, $\mathbf{u}\mathbf{v}^T = (\mathbf{u}\mathbf{v}^T)^T = \mathbf{v}\mathbf{u}^T$. For this equality to hold with $\mathbf{u} \neq \mathbf{0}$ and $\mathbf{v} \neq \mathbf{0}$, the vectors $\mathbf{u}$ and $\mathbf{v}$ must be collinear. That is, $\mathbf{v} = \alpha \mathbf{u}$ for some non-zero scalar $\alpha \in \mathbb{R}$. (If $\mathbf{u}_i \neq 0$, then $(\mathbf{u}\mathbf{v}^T)_{ij} = u_i v_j$ and $(\mathbf{v}\mathbf{u}^T)_{ij} = v_i u_j$. So $u_i v_j = v_i u_j$. This means $v_j/u_j = v_i/u_i = \alpha$ for all j where $u_j \neq 0$. If $u_j = 0$, then $u_i v_j = 0$. If $v_j \neq 0$, then $u_i = 0$ for all i, making $\mathbf{u} = \mathbf{0}$, a contradiction. So if $u_j = 0$, then $v_j = 0$. Thus $\mathbf{v} = \alpha \mathbf{u}$.) Substituting $\mathbf{v} = \alpha \mathbf{u}$ into the expression for $\mathbf{C}$:

$$\mathbf{C} = \mathbf{u}(\alpha \mathbf{u})^T = \alpha \mathbf{u}\mathbf{u}^T.$$

3. Let $\mathbf{w} = \frac{\mathbf{u}}{\|\mathbf{u}\|}$. Then $\mathbf{w}$ is a unit vector (since $\mathbf{u} \neq \mathbf{0}$, so $\|\mathbf{u}\| \neq 0$). This means $\mathbf{u} = \|\mathbf{u}\|\mathbf{w}$. Substituting this into the expression for $\mathbf{C}$:

$$\mathbf{C} = \alpha(\|\mathbf{u}\|\mathbf{w})(\|\mathbf{u}\|\mathbf{w})^T = \alpha(\|\mathbf{u}\|\mathbf{w})(\|\mathbf{u}\|\mathbf{w}^T) = \alpha\|\mathbf{u}\|^2 \mathbf{w}\mathbf{w}^T.$$

   Let $\gamma = \alpha\|\mathbf{u}\|^2$. Since $\alpha \neq 0$ and $\|\mathbf{u}\|^2 > 0$, γ is a non-zero scalar. Thus, we have $\mathbf{C} = \gamma \mathbf{w}\mathbf{w}^T$, where $\mathbf{w}$ is a unit vector and γ is a non-zero scalar.

This completes the proof.

Question 2. Apply the symmetric rank-one quasi-Newton method to solve

$$\min f(x) = \frac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x} - \mathbf{c}^T\mathbf{x}$$

with

$$\mathbf{Q} = \begin{pmatrix} 5 & 2 & 1 \\ 2 & 7 & 3 \\ 1 & 3 & 9 \end{pmatrix} \quad \text{and} \quad \mathbf{c} = \begin{pmatrix} -9 \\ 0 \\ -8 \end{pmatrix}.$$

Initialize the method with $\mathbf{x}_0 = (0, 0, 0)^T$ and $\mathbf{B}_0 = \mathbf{I}$. Use an exact line search.

Solution. To apply the symmetric rank-one quasi-Newton method, I'll follow these steps:

1. Given initial values: $\mathbf{x}_0 = (0, 0, 0)^T$ and $\mathbf{B}_0 = \mathbf{I}$
2. For each iteration k:
   - Compute gradient $\nabla f(\mathbf{x}_k) = \mathbf{Q}\mathbf{x}_k - \mathbf{c}$
   - Compute direction $\mathbf{p}_k = -\mathbf{B}_k^{-1}\nabla f(\mathbf{x}_k)$
   - Determine step size α_k using exact line search. For quadratic functions, the step size is given by $\alpha_k = \frac{-\mathbf{p}_k^T \nabla f(\mathbf{x}_k)}{\mathbf{p}_k^T \mathbf{Q}\mathbf{p}_k}$.
   - Update $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{p}_k$
   - Calculate $\mathbf{s}_k = \mathbf{x}_{k+1} - \mathbf{x}_k$ and $\mathbf{y}_k = \nabla f(\mathbf{x}_{k+1}) - \nabla f(\mathbf{x}_k)$
   - Update $\mathbf{B}_{k+1}$ using the SR1 formula: $\mathbf{B}_{k+1} = \mathbf{B}_k + \frac{(\mathbf{y}_k - \mathbf{B}_k\mathbf{s}_k)(\mathbf{y}_k - \mathbf{B}_k\mathbf{s}_k)^T}{(\mathbf{y}_k - \mathbf{B}_k\mathbf{s}_k)^T\mathbf{s}_k}$

Iteration 1:

- $\nabla f(\mathbf{x}_0) = \mathbf{Q}\mathbf{x}_0 - \mathbf{c} = -\mathbf{c} = (9, 0, 8)^T$
- $\mathbf{p}_0 = -\mathbf{B}_0^{-1}\nabla f(\mathbf{x}_0) = -\mathbf{I}^{-1}(9, 0, 8)^T = (-9, 0, -8)^T$
- Using exact line search: $\alpha_0 \approx 0.1289$
- $\mathbf{x}_1 = \mathbf{x}_0 + \alpha_0\mathbf{p}_0 \approx (-1.1600, 0, -1.0311)^T$
- $\nabla f(\mathbf{x}_1) \approx (2.1689, -5.4133, -2.4400)^T$
- $\mathbf{s}_0 = \mathbf{x}_1 - \mathbf{x}_0 \approx (-1.1600, 0, -1.0311)^T$
- $\mathbf{y}_0 = \nabla f(\mathbf{x}_1) - \nabla f(\mathbf{x}_0) \approx (-6.8311, -5.4133, -10.4400)^T$
- Computing the SR1 update:

$$\mathbf{B}_1 = \mathbf{I} + \frac{(\mathbf{y}_0 - \mathbf{s}_0)(\mathbf{y}_0 - \mathbf{s}_0)^T}{(\mathbf{y}_0 - \mathbf{s}_0)^T\mathbf{s}_0} = \begin{pmatrix} 2.9755 & 1.8857 & 3.2776 \\ 1.8857 & 2.8 & 3.1286 \\ 3.2776 & 3.1286 & 6.4378 \end{pmatrix}$$

Iteration 2:

- $\mathbf{p}_1 = -\mathbf{B}_1^{-1}\nabla f(\mathbf{x}_1)$

- Using exact line search to find α_1
- $\mathbf{x}_2 = \mathbf{x}_1 + \alpha_1 \mathbf{p}_1$
- Compute $\mathbf{s}_1, \mathbf{y}_1$ and update $\mathbf{B}_2$

Question 3. Apply the BFGS quasi-Newton method to solve

$$\min f(x) = \frac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x} - \mathbf{c}^T\mathbf{x}$$

with

$$\mathbf{Q} = \begin{pmatrix} 5 & 2 & 1 \\ 2 & 7 & 3 \\ 1 & 3 & 9 \end{pmatrix} \quad \text{and} \quad \mathbf{c} = \begin{pmatrix} -9 \\ 0 \\ -8 \end{pmatrix}.$$

Initialize the method with $\mathbf{x}_0 = (0, 0, 0)^T$ and $\mathbf{B}_0 = \mathbf{I}$. Use an exact line search.

Solution. Omitted.

Question 4. Let $\mathbf{B}_{k+1}$ be obtained from $\mathbf{B}_k$ using the BFGS update formula. Prove that $\mathbf{B}_{k+1}\mathbf{s}_k = \mathbf{y}_k$.

Solution. The BFGS update formula for $\mathbf{B}_{k+1}$ is given by:

$$\mathbf{B}_{k+1} = \mathbf{B}_k - \frac{\mathbf{B}_k\mathbf{s}_k\mathbf{s}_k^T\mathbf{B}_k^T}{\mathbf{s}_k^T\mathbf{B}_k\mathbf{s}_k} + \frac{\mathbf{y}_k\mathbf{y}_k^T}{\mathbf{y}_k^T\mathbf{s}_k}$$

where $\mathbf{s}_k = \mathbf{x}_{k+1} - \mathbf{x}_k$ and $\mathbf{y}_k = \nabla f(\mathbf{x}_{k+1}) - \nabla f(\mathbf{x}_k)$. We assume that $\mathbf{B}_k$ is a symmetric matrix, i.e., $\mathbf{B}_k^T = \mathbf{B}_k$. The BFGS update preserves symmetry. Thus, the formula can be written as:

$$\mathbf{B}_{k+1} = \mathbf{B}_k - \frac{\mathbf{B}_k\mathbf{s}_k\mathbf{s}_k^T\mathbf{B}_k}{\mathbf{s}_k^T\mathbf{B}_k\mathbf{s}_k} + \frac{\mathbf{y}_k\mathbf{y}_k^T}{\mathbf{y}_k^T\mathbf{s}_k}$$

We want to show that $\mathbf{B}_{k+1}\mathbf{s}_k = \mathbf{y}_k$. We multiply $\mathbf{B}_{k+1}$ by $\mathbf{s}_k$:

$$\begin{aligned} \mathbf{B}_{k+1}\mathbf{s}_k &= \left(\mathbf{B}_k - \frac{\mathbf{B}_k\mathbf{s}_k\mathbf{s}_k^T\mathbf{B}_k}{\mathbf{s}_k^T\mathbf{B}_k\mathbf{s}_k} + \frac{\mathbf{y}_k\mathbf{y}_k^T}{\mathbf{y}_k^T\mathbf{s}_k}\right)\mathbf{s}_k \\ &= \mathbf{B}_k\mathbf{s}_k - \frac{(\mathbf{B}_k\mathbf{s}_k)(\mathbf{s}_k^T\mathbf{B}_k\mathbf{s}_k)}{\mathbf{s}_k^T\mathbf{B}_k\mathbf{s}_k} + \frac{\mathbf{y}_k(\mathbf{y}_k^T\mathbf{s}_k)}{\mathbf{y}_k^T\mathbf{s}_k} \end{aligned}$$

Note that $\mathbf{s}_k^T\mathbf{B}_k\mathbf{s}_k$ is a scalar and $\mathbf{y}_k^T\mathbf{s}_k$ is also a scalar. Assuming $\mathbf{s}_k^T\mathbf{B}_k\mathbf{s}_k \neq 0$ and $\mathbf{y}_k^T\mathbf{s}_k \neq 0$ (the latter is a standard condition for the BFGS update, often $\mathbf{y}_k^T\mathbf{s}_k > 0$ is required to maintain positive definiteness), we can simplify the expression:

$$\begin{aligned} \mathbf{B}_{k+1}\mathbf{s}_k &= \mathbf{B}_k\mathbf{s}_k - \mathbf{B}_k\mathbf{s}_k + \mathbf{y}_k \\ &= \mathbf{y}_k \end{aligned}$$

Thus, we have shown that $\mathbf{B}_{k+1}\mathbf{s}_k = \mathbf{y}_k$.